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How does change in CEOs' strategic orientations in their social media communication impact firm performance during crisis? A longitudinal study

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ABSTRACT

The social media (SM) presence of the CEOs and their sustainability-related communication plays a major role in handling business communication and shaping firm strategic outcomes during a crisis. In turbulent times of crisis, stakeholders seek credible information for their better decision-making towards firms. Therefore, it becomes essential for the CEOs to engage with the stakeholders and manage firm value strategically. This empirical study utilizes the Twitter posts of Fortune 1000 CEOs for their sustainability communication through their different orientations i.e., SDG, entrepreneurial, and leadership styles. The study employs mixed research methodology through a longitudinal event study to determine the change in the CEOs' strategic orientations, its influence on the CEOs' SM reputation, and firm performance before and during the crisis. Findings showcase that the change in CEOs' strategic orientation in their sustainability-related disclosures positively influences stakeholder engagement and the CEOs' SM reputation during a crisis. Also, the strategic orientations of the CEOs and their SM reputation as a resource provide competitive advantages to the firms for their enhanced firm financial performance. Our study contributes to both research and practice of using SM strategically for their sustainability communication to achieve improved firm performance during a crisis.

1. Introduction

The global crisis of the COVID-19 pandemic since February 2020 led to an influx of news from unreliable sources, leading to an increase in the phenomenon of fake news on SM. According to the MIT Technology Review, the first SM "information epidemic" befell during the outbreak of COVID-19, which was further declared as an "infodemic" by the World Health Organization (WHO) (Hao and Basu, 2020). It had hit the businesses and severely affected them in uncertain ways propelling them to rethink and redesign their business communication processes (Li et al., 2021).

As brand ambassadors, active and responsible participation of CEOs in public and private networks to address the global challenges of crisis is a need of the hour. Timely communication and engagement with the stakeholders are crucial to avoid loss of trust in information shared by organizations and economic impacts (Banks et al., 2022). For gaining business value while connecting with different stakeholders, certain SM strategies are desired for populating engaging content that is of interest to the community (Culnan et al., 2010). However, there is a paucity of literature on whether the CEO's orientations in their communication strategy with the stakeholder community change and how this change creates business value (Jha and Verma, 2022; Samimi et al., 2022). To bridge this theoretical silo, we consider different orientations, i.e., SDG orientation, entrepreneurial orientation (EO), and leadership style of the CEOs in their sustainability-related communication on SM platform and explore whether the CEO's orientation changes during a crisis through the following research question.

RQ1: How do CEOs' different orientations on their SM sustainability discourse evolve during the COVID-19 pandemic global crisis?

SM platforms being more dialogic and interactive instruments are paramount for firms to develop relations with the audience quicker during crises than other classic media platforms such as websites or blogs (Schultz et al., 2011). However, in such an emergency scenario, quality communication and engagement through credible sources become crucial to avoid loss of trust, and build reputation (Viola et al., 2021). SM use has, therefore, developed as a potent medium for CEOs

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for their novel strategic communication and stakeholders for their informed decision-making (Elliott et al., 2018). Communication of business strategies and actions surrounding sustainability helps improve stakeholder engagement and shapes the firm-stakeholder relationship (Rosamartina et al., 2022; Men and Tsai, 2016). The different orientations of the firm's CEO in their communication influences the perception of leadership capabilities (Heavey et al., 2020), enhances stakeholder engagement and, hence, their reputation (De Luca et al., 2022). Therefore, through the following research question we investigate the change in association of CEOs' different orientations in their SM communication and their SM reputation during a crisis.

RQ2: Does change in the evolution of the CEOs' orientations towards SDG, entrepreneurship, and their leadership style impact their social media reputation during the COVID-19 pandemic crisis?

The value derived from SM resources hinges on the firm's willingness and ability to utilize these resources efficiently. Thus, it becomes crucial for the firm's use of SM to achieve a higher reputation and firm value (Liu et al., 2023; Wibowo et al., 2022; Bergh et al., 2010). In this study, we posit that SM reputation built from different orientations of CEO in their communication contributes to building a competitive, sustainable advantage for the firms and influences firm performance. The present study, therefore, explores how reputation built from different orientations of CEO in their SM communication contributes to building competitive, sustainable advantage for the firms and influences firm performance. To investigate this further, we propose the following research question.

RQ3: Does change in the SM reputation of CEOs due to the outbreak of the COVID-19 pandemic crisis influence the firm's financial performance?

A need for the firms and their CEOs in their decisions about strategy and communication towards sustainable development goals (SDG) adoption has been highlighted as crucial for economic and non-economic gains (Aguilera et al., 2021). Muñoz and Cohen (2017) also emphasize adopting sustainability-oriented practices through engagements with stakeholders to establish collective visions and values between the firm and the stakeholder groups. Therefore, to conduct this study we consider sustainability-related tweets of Fortune 1000 CEOs. We utilize computer-aided text analysis, Natural Language Processing (NLP), and econometric modeling methods for longitudinal data before and during the pandemic.

Our study contributes to the research by confirming that crisis motivates CEOs to effectively utilize SM for stakeholder engagement, to influence their reputation and firm performance. This event study uses stakeholder theory and RBV theory as theoretical lens to explain (1) the effective utilization of the SM platforms for sustainability disclosure during a crisis, (2) how the change in CEO's different orientations i.e. SDG, entrepreneurial, and leadership style while sharing particular content on SM influences stakeholder engagement levels and CEO reputation on SM, and (3) how CEO's SM reputation built from the adoption of different strategic orientations in their SM communication act as resource for the firm and has implications on firm performance during the crisis. The intriguing empirical results verify the change in CEO strategic orientations while discussing sustainability initiatives on SM, its influence on CEO-stakeholder engagement, and CEO SM reputation. Moreover, the change in reputation built from strategic change in CEO orientations plays a significant positive role in shaping firm performance during the crisis.

The remaining paper is structured as follows: The next section provides the past literature of the use of SM during crisis and for sustainability disclosures. The third section presents the theoretical background and hypotheses formulation. Next, we discuss the methodology used to conduct the study followed by the analysis results. Finally, the study's theoretical, practical implications, limitations and future scope followed by the conclusion are discussed.

2. Literature review

2.1. Social media use during crisis

Any crisis, financial, economic, health, pandemic, or natural disaster occurring locally or globally causes abrupt effects on business operations and firm profitability. During a crisis, time pressure (from the media, public, and employees) and uncertainty influence the public relations of the firms with the stakeholders (Van der Meer et al., 2017). It is crucial for a firm to pay attention to communication style and effective response strategy for addressing the public during crisis situations to avoid unfavorable impressions (Turan, 2022). Digital innovation and the role of information technologies, especially SM provide huge potential to address the challenges during and post-crisis through effective communication by the firms and help them to strategically adapt to the changing environment (Buck et al., 2022).

During societal crises, the public looks for relevant information from multiple sources and has concerns about the credibility of the information for better decision-making (Oh et al., 2013). In such a situation, SM adoption by strategic leaders as a medium for novel interactions and influencing the stakeholders through their engagement behaviors and strategies have become popular and a widely accepted practice (Heavey et al., 2020; Li et al., 2022). In the past, narratives from CEOs have been linked to increased organizational credibility and legitimacy during times of crisis (Fowler, 2017). Moreover, the adoption and utilization of SM during the pandemic have been proven to improve business performance and create sustainable competitive advantages (Akpan et al., 2022). The benefits of utilizing SM during a crisis highly encourage firms to information sharing and stakeholder engagement practices.

2.2. Social media and sustainability disclosures

Firms use SM strategically for propagating information and public engagement with different agendas including sustainability initiatives (Cho and Roberts, 2010). Previous studies throw light on the category of firms active on SM, popularly communicated content, time and situation when sustainability disclosures influence public decision-making, and SM as a consumer's choice to interact and engage with the firms (Becker-Olsen et al., 2006; Amin et al., 2021; Saxton et al., 2019). Leveraging SM allows firms and their leaders to articulate their commitment, communicate their strategic decisions, and respond to stakeholder inquiries surrounding their sustainability initiatives (Men and Tsai, 2016). These sustainability disclosure on SM are the reflection of their sustainability efforts and influence consumer's decision-making (Jha and Verma, 2022), and if not monitored or strategized in a direction can lead to transparency and credibility challenges for the firms (Dobija et al., 2023). Therefore, firms and their CEOs must strategically implement effective communication practices on SM for sustainability disclosures.

A recent editorial by Gruber (2023) highlights the pressing need for organizations to operate more sustainably, requiring leaders to be inclusive and collaborative in their decision-making in developing business strategies and activities. Khizar et al. (2022) also emphasize the limitations in the prior research concerning the complementary effects of firms' and their leader's sustainability orientation alongside other strategic orientations, such as entrepreneurial orientations, the organization's environmental conditions, and stakeholder engagements, on performance. Organization's adoption of sustainable practices in their business operations has been in existence since the proposal of SDG by the United Nations in 2015 (Nations, 2015), but this adoption of sustainable agendas gained momentum following the onset of the COVID-19 pandemic. CEOs with entrepreneurial orientation are expected to allocate the company's resources towards corporate philanthropy that may generate a good reputation for them (Zhang et al., 2021). Also, strategic leaders' interaction influences social media engagements that further have implications on decision-making and firm outcomes (Heavey et al., 2020). Previous studies showcase evidences of the

relationships between different firm orientations, and strategic resources on the firm performance (Colton et al., 2010) but there is still a succinct gap in the literature on how strategic disclosures change during a crisis and have further implications on firm performance. In this study we explore the implications of CEO possessing different strategic orientations i.e. sustainability, entrepreneurial and leadership styles in their social media communication during any crisis. The following section discusses the theoretical background and hypotheses formulation to address the abovementioned gaps in the literature.

3. Theoretical background and hypotheses development

3.1. CEO's orientation in strategic communication on social media during crisis

Firms use different strategies for disseminating information and shaping up an overall information environment on SM (Jung et al., 2018). Since the narratives from the CEOs have been associated with enhanced organization's credibility and legitimacy during the period of crisis (Fowler, 2017), it is crucial for the CEOs to adopt a unique strategic approach by communicating on topics that are pertinent at the time and cognizant of stakeholders' perspectives to cultivate meaningful engagements (Men and Tsai, 2016). The different orientations of the CEO i.e. CSR orientation, entrepreneurial orientation, sustainability orientation, leadership styles and many more in their communication related to various topics have been extensively examined in the literature, impacting business performance and organizational success (Clark et al., 2023; Gruber, 2023; Xia et al., 2023).

Recent studies have posited sustainability orientation as one of the important strategic orientations of the firms and their leaders (Khizar et al., 2022; Gruber, 2023). SDG orientation of the CEOs helps the organizations to address the challenges due to COVID-19 pandemic and showcases the organization's willingness to strategize their business practices, that has implications on performance and contribute to the society (Thore, 2022; Aguilera et al., 2021). SDG orientation of the firms hence caters to the informational demands of the stakeholders and yield long-term organizational benefits by integrating sustainability principles in their business (Khizar et al., 2022). Therefore, it is crucial to identify if there is any change in CEO SDG orientation in their social media communication before and during the crisis. To determine this we state the following hypothesis H1a. In our study, SDG orientation in CEO communication on SM involves measuring of how much a CEO talks about 17 SDGs proposed by the United Nations, for example Zero hunger, Poverty, Climate action, Education etc. (Nations, 2015).

H1a. There is a significant change in SDG orientation of CEOs on SM during the COVID-19 pandemic crisis.

With the downfall in business growth witnessed during the pandemic, entrepreneurial activities by the firms and their leaders reflect the firm's decision-making practices. Literature shows evidence on the increasing importance of entrepreneurial corporate activities in creating sustainable development in organizations with the aim of influencing market conditions (Schaltegger and Wagner, 2011). EO as an organizational attribute reflects the entrepreneurial capabilities of businesses through risk-taking, proactiveness, and innovativeness (Covin and Slevin, 1989; Miller and Le Breton-Miller, 2011). Recent literature also suggests that the CEOs EO at an individual level has diverse implications on their firm competitiveness and performance outcomes (Clark et al., 2023). Different typology of entrepreneurial communication adopted by CEOs over various platforms helps mitigate audience uncertainty about their entrepreneurial decisions and evoke affirmation of a firm's distinctiveness in their strategy (Boling et al., 2016; Fischer and Reuber, 2014). Roxas (2021) argues that firms achieve greater success when their strategic decision-makers possess an EO, enabling them to capitalize on new opportunities that emerge from engaging in sustainability initiatives despite limited resources. Therefore, we state the hypothesis H1b as follows. In our study, EO is the ability of the CEO with his/ her different entrepreneurial capabilities has the ability to manage the social media communication to build reputation among stakeholder community (Zhang et al., 2021), and its measures include innovativeness, proactiveness, risk-taking, and autonomy (Covin and Wales, 2012).

H1b. There is a significant change in the EO of CEOs on SM during the COVID-19 pandemic crisis.

The need for cutting-edge, real-time data and the spread of specialized knowledge that fosters personalized connections and increased social impact encourages leaders to be more attentive to the informational need and concerns of their stakeholders (Avolio et al., 2000). In the literature, it is well proven that the top executives' leadership traits contribute to the firm's strategy formation and performance (Hambrick and Finkelstein, 1987). Different leadership styles possessed by the leader have varied outcomes in terms of how they are perceived by the audience and how it affects their performance (Dinh et al., 2014). Through their strategic leadership styles in their narratives, they try to overcome the information asymmetry among the stakeholders (Shane and Cable, 2002). Evidence in past literature showcases the important role of responsible leaders during crisis times in preventing information asymmetry among stakeholders (Varma, 2021). However, there is a mixed argument between the impact of strong leadership, and accuracy and objectivity on firm performance during the crisis (Kraus et al., 2020). Therefore, through the following hypothesis, in this study, we identify if there is any change in the leadership style, i.e. transactional or transformational, of the CEOs as a strategic move to address the challenges of information asymmetry due to the COVID-19 pandemic.

H1c. There is a significant change in the leadership style of CEOs in SM during the COVID-19 pandemic crisis.

3.2. CEO's orientation in strategic communication and their reputation on social media

Firms and their CEOs disclose and communicate with the SM stakeholder community about their business strategies and actions surrounding sustainability so as to improve stakeholder engagement and build firm's reputation (Nasr et al., 2022; Rosamartina et al., 2022). Stakeholder theory provides a perspective on how stakeholders influence firm behavior and how firms make strategic decisions about communication strategies to fulfil stakeholder demands (Freeman, 1984). The increased scope of SM for two-way communication among the stakeholder community leads to increased power of the stakeholders, having higher engagement and financial performance benefits (Kim and Youm, 2017). Higher CEO-public engagement in SM has also proven to positively influence public trust and satisfaction, hence directly influencing the quality of firm-stakeholder relationships (Men and Tsai, 2016).

A strategic leader's engagement on SM is characterized as a set of behaviors aimed at effectively utilizing SM capabilities to communicate with the stakeholders, thereby facilitating the development and execution of firm's strategies (Heavey et al., 2020). Literature shows evidence that the level of stakeholder engagement on the messages posted by firms on SM surrounding SDGs is highly influenced by factors such as content type, hashtag use, use of links, length of the post, the industry type, and country (De Luca et al., 2022; Lee et al., 2018). Quality and timeliness of the SDG disclosure on SM by the firms influence the intensity of stakeholder engagement activities (Pizzi et al., 2021). In addition, CEOs with effective and strategic EO leverage SM to build reputation and create value via innovation and strategic alignment with the stakeholders (Hitt et al., 2001) and has performance implications (Saiyed et al., 2023; Bauweraerts et al., 2023). Similarly, the leadership styles of CEOs are demonstrated in their communication on varied media platforms help them to build organizational identity (Kuhn,

2008), and enhance performance (Bormann and Rowold, 2018). The interpersonal communication of the CEOs with perceived authenticity helps in building positive public engagement, trust, and satisfaction of the stakeholders (Men and Tsai, 2016; Hall et al., 2004). It is assumed that CEOs of the firm through their different orientations in their communication shape up the public opinion about them and their firms. Their orientation in the media narrative affects the perception of their leadership capabilities, which contribute to enhanced reputation (Heavey et al., 2020). Evidence in past literature also highlight the important role of responsible leaders during the crisis times (Varma, 2021).

Yao (2022) recommends that firms change their strategies for their SM communication regarding the crisis to build a positive engagement with the consumer for crisis management. Therefore, through the following hypotheses, stakeholder theory as a theoretical lens has been used to investigate whether CEO's different orientations (SDG orientation, EO, and leadership style) in their SM communication about sustainability have any implications on their SM reputation during a crisis.

H2a. There is a positive change in the association of CEO's SDG orientation and their SM reputation during the COVID-19 pandemic crisis.

H2b. There is a positive change in the association of CEO's EO and their SM reputation during the COVID-19 pandemic crisis.

H2c. There is a positive change in the association of CEO's leadership style and their SM reputation during the COVID-19 pandemic crisis.

3.3. CEO's social media reputation and firm financial performance

The firm's SM strategy increases the level of engagement with stakeholders by instilling a sense of trust and collaboration among the audience (Mention et al., 2019). Previous literature presents evidence on mixed performance outcomes regarding SM engagement and the reputation of the CEOs on the firm performance. A vast body of research brings out the positive aspects of the CEO's engagement on SM while building trust and a long-term orientation, influencing investor behavior and perceptions, and subsequent impacts on the firm financial performance (Flammer and Bansal, 2017). Some studies showcase the outcome of CEO's negative SM reputation on the decrease in the firm equity, leading to financial risks (Luo et al., 2013). There is evidence that demonstrates the CEOs' use of SM to manage the crisis and mitigate the negative financial impacts (Pfarrer et al., 2008).

Lam et al. (2019) emphasize that value creation of the SM resources depends on the firm's motivation and capabilities to make use of these resources. Therefore, it is crucial for the firms to allocate resources and adopt the strategies of SM use to achieve better reputation and performance (Dwivedi et al., 2021). Also, Bergh et al. (2010) highlight reputation as a strategic resource that is difficult to imitate by other organizations. Therefore, the reputation of the CEOs on SM build through the different inimitable strategic orientations plays a major role in value creation and firm performance.

Enhancing reputation requires leaders to wisely nurture interdependencies and complex relationships among various strategies, hence it is considered as an intangible strategic asset and valuable resource that has financial consequences on the firm (Boyd et al., 2010). SM reputation of the CEOs has far-reaching implications for the firm financial performance through effective engagement and strategic communication. CEOs connectedness and higher stakeholder engagement affect the organizational future performance (Horton et al., 2012; Liu et al., 2023).

Therefore, through the lens of RBV theory, authors explore the relevance of CEOs' different strategic orientations and their reputation as a unique strategic resource in their sustainability-related SM communication during the crisis and its impact on firm financial performance. RBV theory explains the firm's competitive advantage by aligning its resources that comply with VRIN i.e., valuable, rare, inimitable, and non-substitutable (Colbert, 2004). According to Barney (2001), not all resources can provide firms with a competitive advantage, as different capabilities and resources have varying effects on a firm's ability to achieve differentiated advantages. The following hypotheses investigates the impact of the CEO SM reputation on the firm financial performance during the COVID-19 pandemic crisis.

H3a. There is a positive change in the relationship between CEOs' social media reputation and firms' return on assets (RoA) during the COVID-19 pandemic crisis.

H3b. There is a positive change in the relationship between CEOs' social media reputation and firms' sales revenue during the COVID-19 pandemic crisis.

H3c. There is a positive change in the relationship between CEOs' social media reputation and firms' TobinsQ ratio during the COVID-19 pandemic crisis.

The research framework used for the study is highlighted in Fig. 1.

4. Methodology

An inductive approach suggested by Berente et al. (2019) and Kar and Dwivedi (2020), has been adopted to study the sustainability communication of CEOs on Twitter. To conduct this study, we have employed mixed research methodology, which provides a better understanding of complex phenomena by integrating qualitative and quantitative methods (Molina-Azorin, 2012) around the SM content for analyzing social relations and the embeddedness of social actions in firms. The current study combines social media analytics using natural language processing (NLP) to build econometric models through an event study over longitudinal data collected before and during the pandemic.

4.1. Data collection and pre-processing

The current empirical study uses the user-generated content of Fortune 1000 CEOs on SM platforms. Twitter is CEOs' most widely used SM platform for information sharing and managing public relations (Matthews et al., 2022). Therefore, we checked for the presence of CEOs on Twitter using Google search and the Twitter platform. Only 15 % out of Fortune 1000 CEOs were found to have their profiles on Twitter. Thereafter, the Twitter profile timeline of the CEOs having verified accounts were extracted through text mining methods using Twitter API (Application Programming Interface) with Python (Russell, 2013). The information retrieved from the user timeline includes text messages posted, date when the post was created, number of times the post was liked by others, number of times the post was shared by others, the number of user followers, etc. which were documented in the CSV files for individual CEOs.

The tweets created by the CEOs on Twitter are in an unstructured format. Therefore, it was important to pre-process and clean the data fetched from CEOs' Twitter timeline before analysis. For this purpose, tweets were first converted into lower case followed by stemming and lemmatization of the text. Further, URLs, hypertexts, hashtags, numerical values, punctuation, and extra spaces were removed. Also, a list of stop-words was created to remove keywords that add no value to the data before further analysis.

In this event study using longitudinal data, we identify the change in the CEOs' strategic orientations in their SM communication due to COVID-19 pandemic crisis, we first segregate the tweets during the timeframe of the years 2018 to 2021, resulting in a total of 13,522 tweets. Our study focuses on the sustainability-related communication of the CEOs, tweets discussing different SDGs as sustainability agenda were then extracted using the dictionary method of automated content analysis (ACA). Content analysis is a computer-aided text analysis

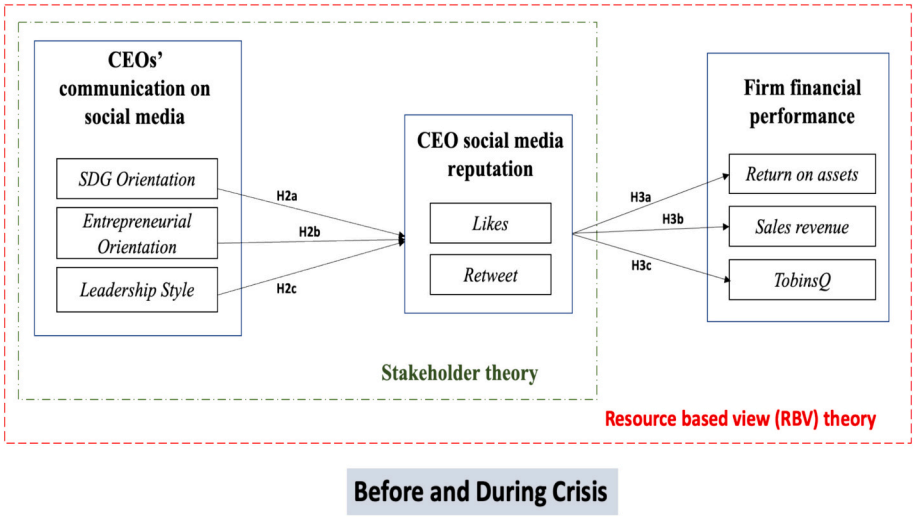


Fig. 1. Research framework for the study.

method often used to analyze organizational narratives (Short et al., 2010). The dictionary method is an efficient content analysis method that can be customized and minimizes human bias (Grimmer and Stewart, 2013; Krippendorff, 2018). A total of 8463 tweets discussing sustainability were derived. The sustainability tweets for 2018 and 2019 were considered for the "before crisis phase" and sustainability tweets for 2020 and 2021 for the "during crisis phase". Different engagement levels of the CEOs in their SDG tweets were observed. Therefore, to control this bias, a threshold of a minimum of 15 tweets from each CEO that discussed sustainability was considered for the specified time period. Only 8 % of the Fortune 1000 CEOs were selected for further analysis.

The financial data for the selected firms were derived from the Bloomberg terminal, only 6.7 % were selected for final analysis with the availability of all the data points. To examine the leading effects of CEO's different strategic orientations in their SM communication on financial performance, we gathered financial data of the firms spanning the years 2018 to 2022.

The study sample includes companies belonging to different sectors. The majority of the sample firms belong to the technology sector i.e. 35 %, 10 % belong to the financial sector, while business services; retailing; transportation; materials, engineering and construction include 6 % each, healthcare; energy; and Industries include around 5 % each, whereas media; household; hotels and restaurants; and motor vehicles include around 3 % each, rest wholesalers; real estate; and chemicals include around 2 % each of total sample. Furthermore, the CEOs of the sample firms are mostly males with 96 % being males and 4 % being females.

4.2. Operationalization of variables

4.2.1. Measures for CEO's different orientations

ACA through dictionary method and bag-of-words was employed to measure the different strategic orientations of the CEOs on SM. The ACA technique for supervised learning helps avoid human error while categorizing the different CEO orientations in their Twitter posts. The qualitative text was analyzed and interpreted in quantitative form by tracking the linguistic patterns and categorizing tweets depending on the frequency of word patterns of occurrences (Krippendorff, 2018). For measuring different strategic orientations of the CEOs in their SM disclosures, already well-established scales were referred to create the codebooks of a dictionary of keywords belonging to each orientation type.

The triple bottom approach using economic, environmental, and social dimensions mapped with 17 SDGs and relevant keywords for each dimension were extracted using the guidelines proposed by the United Nations surrounding 169 indicators (Nations, 2015). The EO measurement model by Covin and Wales (2012) has been used for measuring the four dimensions of EO i.e., innovativeness, proactiveness, risk-taking, and autonomy. The leadership style of the CEOs was measured as transactional and transformational types, and the widely accepted MLQ (Multi-factor leadership questionnaire) instrument has been used for measuring the leadership style possessed by the CEO on SM (Tejeda et al., 2001).

The codebooks, consisting of a dictionary of keywords for each strategic orientation, were prepared using the above-mentioned scales. To prepare the bag of words, ten researchers were appointed, four with a research background in the field of sustainability, three from the area of entrepreneurship, and three from the human resource area working on leadership research. Each of these experts had at least 10 years of experience in their respective fields in academia and industry. These experts were provided with a succinct background, the purpose of conducting this study, and the scales for each CEO orientation type. The responses were then combined to form a single codebook for different orientations possessed by CEOs on SM. Reliability and validity tests were performed before finalizing the dictionary codebook for further analysis.

Further, supervised ACA is applied to the CEOs' posts for the presence of SDG orientation, EO, and Leadership style by using the custom dictionary codebooks. In the next step, the quantitative scores for the presence of each strategic orientation were derived. Using a human coding mechanism, a relative scoring of each orientation was done with absolute scoring between 0 (low presence of the defined individual categories) and 5 (high presence of the defined individual categories). The leadership style had been used as a dummy variable; therefore, we further divided it into the categories as follows: a score of "0" was given for a CEO with a transactional leadership style, a score of "1" for a CEO with a transformational style, and a score of "2" when a CEO possesses both the leadership qualities. Finally, the quantitative values against SDG orientation, EO, and leadership style for each Fortune CEO were derived for further statistical modeling.

4.2.2. Measures for CEO social media reputation

Engagement in SM helps assess the effectiveness and success of the SM management strategies of the firms (Jiang et al., 2016). SM engagement is a multidimensional concept that involves different engagement categories identified by Dessart et al. (2015) including affective, cognitive, and behavioral. For this study, we consider the behavioral perspective for measuring the degree of SM engagement of

CEOs while discussing sustainability. Liking, retweeting, commenting, and clicking are popular measures for SM user engagement (Lee et al., 2018). Sharing is driven by the content created on SM that strongly represents the exchange of ideas, interesting content, or experiences, whereas endorsing is the act of recommending specific content in a group that can also be referred to as advocating specific brands or organizations. Liking and sharing the content created by others showcases positive and moderate active engagement behaviors on SM (Dolan et al., 2019). However, the engagement of the stakeholders in the CEO's posts helps them build a reputation on SM (De Luca et al., 2022; Grover et al., 2019). Hence, behavioral measures of the CEO's SM reputation are taken as an endorsement (favourite count/ likes on the tweet), and advocacy (retweet count). In our study, we take an average of both number of likes and number of retweets on a CEO tweet as a measure of the CEO's SM reputation.

4.2.3. Measures for firm performance

To explore the effects of CEO's SM reputation on different measures of firm financial performance, we used return on assets (RoA) and sales revenue, which are usually defined as accounting measures of performance (Barton et al., 2010; Tubbs, 2007), and TobinsQ, which is widely used market variable (O'Sullivan and McCallig, 2012). These are commonly used in management and financial research when evaluating the impact of sustainable practices within business organizations (Das-Gupta, 2022; Nirino et al., 2021). Table 1 presents the significance of each financial performance indicator.

4.2.4. Control variables

The number of users participating in content sharing on Twitter is influenced by the number of followers associated with a Twitter handle. Twitter users with more followers get likes and shares of their posts more frequently (Shi et al., 2014). Similarly, the firm's performance is influenced by the firm size measured as the employee count in the organization (Benitez-Amado and Walczuch, 2012). Also, a high variation was found in the follower count and the firm size across the study sample. Since, taking a natural log of the variable helps to overcome the skewness in the data and provides a more approximately normal distribution of the variables (Benoit, 2011). Therefore, to avoid bias, the natural log of the user follower count and the natural log of the employee count of the firm were used as control variables. Different sectors may have different strategies towards sustainability and the pandemic may have a varying effect on firms belonging to different sectors (Li et al., 2021). Therefore, to avoid bias in our results, we control for the sectors in this study.

4.3. Reliability and validity

Custom dictionaries prepared for measuring the SDG orientation, EO, and Leadership style possessed by CEOs on SM were further checked for reliability and validity through the inter-coder mechanism using Cohen's kappa method (Krippendorff, 2004). The panel consisted of six members from industry and academics working in sustainability, social sciences, entrepreneurship, and human resources, with an average experience of 14 years. A brief about the references for codebook generation was provided to the panel members so that they are in sync with the background of the scales or models adopted by the authors while preparing the dictionary. Firstly, three groups of two members each were formed to validate each of the three codebooks. Secondly, the codebooks were circulated to different groups for the second round of validation. Finally, all members mapped the keywords to the codebooks with a common consensus of 0.94, 0.90, and 0.89 for SDG orientation, EO, and leadership style respectively.

Furthermore, we also conducted the Fliess Kappa test in extension to the Cohen's Kappa test. Fliess kappa test for inter-rater reliability is an efficient method for establishing consensus among two or more researchers coding qualitative data to achieve a common consensus in psychology, sociology, and communication (Fleiss, 1971; McDonald et al., 2019). For this purpose, we added three more experts from the academics working in either sustainability, entrepreneurship, or human resources with a minimum of 7 years of experience. We then seek a response from one additional expert for each of the strategic orientations of all the CEOs in the study sample. Combining the results of the coding verification from three experts for all the CEOs' strategic orientations we reached the consensus of 0.96, 0.90, and 0.92 for SDG orientation, EO, and leadership style respectively.

Table 1
Bloomberg variable description of firm financial performance indicators.

Financial performance indicators	Bloomberg definitions
Return on assets	Profitability of a company relative to its total assets (in percentage)
Sales revenue	The gross revenue from any operating activity. It is the total sales generated after the deducting of allowances, discounts, sales returns, and sales-based taxes
TobinsQ	Firm's market value divided by replacement value of its assets

4.4. Data analysis and model specification

This empirical study analyzes the longitudinal data using a mixed research methodology from computational social science and computer science in three phases. Firstly, the content analysis of the qualitative SM content posted by CEOs was analyzed using the dictionary method of natural language processing (NLP) for ACA. Second, the quantitative scores for CEOs' different strategic orientations were derived using a human coding mechanism. Lastly, multiple linear regression methods of econometric modeling have been used to measure the hypotheses.

The change in statistical significance of the SDG orientation, EO, and leadership styles before and during the COVID-19 crisis has been conducted. For deciding between the parametric and non-parametric tests of statistical significance testing, the normality and homogeneity of the variables were tested using the Kolmogorov-Smirnov test and Levene's test, respectively. The findings show that the variables were neither normal nor completely homogenous. Parametric tests were rejected because of the above two conditions, hence considering the type of variable Mann- Whitney statistical test of non-parametric tests was adopted for checking statistical significance between the two groups (before and during the COVID-19 pandemic) for testing the hypotheses H1a, H1b, and H1c. Multiple regression analysis was applied to determine the impact of the CEO's different orientations on their SM reputation and its further implications on firm financial performance before and during the COVID-19 pandemic crisis. The following econometric models have been used to conduct the study:

Model specification for capturing the relationship between CEO orientation and social media reputation:

SM reputation for a CEO of a firm 'i' in time period 't' can be obtained as:

$$CEO_SM_Reputation_{it} = \alpha + \beta_0\, SDG_{it} + \beta_1 EO_{it} + \beta_2 Leadership\ style_{it} + \beta_3 Follower\ count_i + \varepsilon_{it} \quad (1)$$

Model specification for capturing the relationship between SM reputation of CEOs and firm performance:

The effect of the CEO's SM reputation on the firm performance at a given time period 't' is obtained as:

$$RoA_{i(t+1)} = \alpha + \beta_4\, CEO_SM_Reputation_{it} + \beta_5 ln(No.of\ emp.)_{it} + \beta_6 Sector_i + \varepsilon_{it} \quad (2)$$

$$Sales_Revenue_{i(t+1)} = \alpha + \beta_6\ CEO_SM_Reputation_{it} + \beta_7 ln(No.of\ emp.)_{it} + \beta_8 Sector_i + \varepsilon_{it} \quad (3)$$

$$TobinsQ_{it} = \alpha + \beta_{12}\ CEO_SM_Reputation_{it} + \beta_{19} ln(No.of\ emp.)_{it} + \beta_{20} Sector_i + \varepsilon_{it} \quad (4)$$

Note: $t = 1$ (before the COVID-19 pandemic, Years- 2018 and 2019), 2 (during the COVID-19 pandemic, Years- 2020 and 2021).

Over a given period, the size and speed with which SM message diffuses depend on multiple dimensions such as message fluency, diffusion strategy adopted, content type, source type, network strength, etc. (Zhu et al., 2020). The propagation of this influence is measured through the cascade of likes and retweets, which may involve a time delay (Chen et al., 2012). Horton et al. (2012) highlight the association of a firm's connectedness not only with its current performance but also with its future performance. Therefore, considering the propagation delays in the information diffusion, the authors feel that the CEOs' SM activity cannot be observed spontaneously on the firm performance, hence authors considered the lead effect of the tweet propagation influence on the firm financial performance. Hence, endorsement and advocacy in CEO tweets in time t are regressed with financial performance indicators of time t + 1.

5. Findings

Out of Fortune 1000 CEOs, only 15 % of the CEOs were found to be present on Twitter. After scraping the timeline of these CEOs, it was found that only 8 % of the CEOs were actively participating in SM communication for engaging with Twitter users. The CEOs who did not post in either of the two-time frames before and during the pandemic were excluded from the study. The firms for which the financial data was unavailable for any of the above-mentioned periods were also not considered, hence for the final analysis purpose, only 6.7 % of the CEOs were considered.

5.1. Descriptive findings

From the CEOs tweets, it is evident that their orientation towards different sustainability types is evolving in their SM narratives with time and a change in pattern has been observed from discussing economic aspects more before the pandemic, while they discuss more about the social and environmental aspects of sustainability after onset of the COVID-19 pandemic crisis.

Fig. 2 showcases the evolution of the CEOs' sustainability disclosures on Twitter while addressing the online audiences to navigate the COVID-19 pandemic crisis. The timeline evolution of the CEOs' sustainability discussions demonstrates that topics surrounding equality, cybersecurity, and corporate are spread throughout the timeframe of 2018 to 2021 with similar intensity. The discussion surrounding vision, celebration, kind, study, and disease follows a discontinuous pattern starting from 2018, 2019, and then 2021. CEOs' discussions surrounding covid, pandemic, crisis, health, decision, and climate are evident in late 2019 and 2020. Topics related to entrepreneurship, startups, trinet and education are established in the CEOs' disclosure from 2019 to 2021. The above-mentioned topics demonstrate that CEO discussions surrounding the economic and social SDGs are spread across pre and post-pandemic time. Whereas topics related to environmental SDG dimensions are discussed more during the pandemic. Thus, it can be inferred that there is a change in CEOs' strategic orientation towards different sustainability dimensions during the crisis.

5.2. Effect of the pandemic on different strategic orientations of CEOs on social media

Mann-Whitney *U* tests have been performed to check for the shift of CEOs' strategic orientations towards SDGs, EO, and leadership styles after the manifestation of the pandemic. When checked for SDGs, EO, and leadership styles possessed by CEOs on SM before and during the pandemic, the effect sizes were found to be 0.317, 0.131, and 0.279 respectively. According to Cohen's criteria of effect sizes, SDG orientation exhibits a medium effect, EO exhibits a small effect, and leadership styles exhibit a nearly medium effect due to the onset of the pandemic (Cohen, 1992). Hence, the hypotheses H1a and H1c are accepted, and H1b is rejected.

5.3. Event study 1: CEOs' strategic orientations and social media reputation

Multiple linear regression has been used to identify the change in the association between the different orientations of CEOs towards SDGs, entrepreneurship and leadership style, and their SM reputation due to

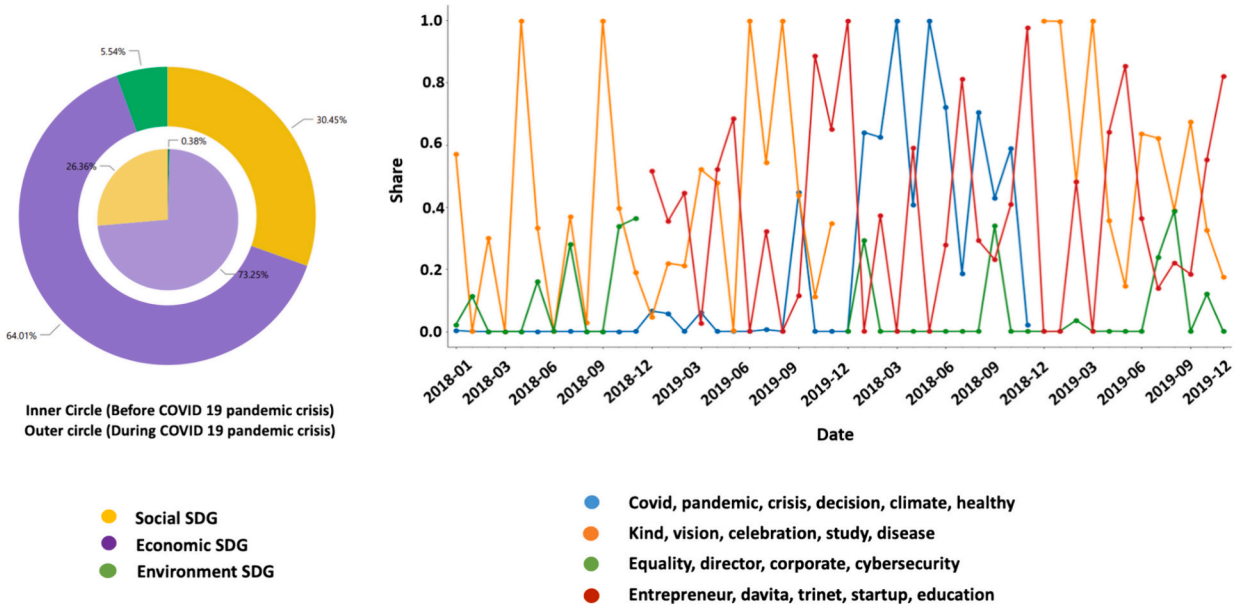


Fig. 2. Change in CEOs discussions and timeline evolution surrounding SDGs before and during the pandemic.

Table 2
Relationship between CEO's orientations and their social media reputation.

Independent variables	Dependent variable	
	CEO_SM_Reputation	
	Before-crisis	During-crisis
SDG Orientation	0.413***	0.283***
Entrepreneurial Orientation (EO)	0.097	0.196***
Leadership Style	0.209	−0.098
Control Variables		
Follower Count	Controlled	
R^2	0.155***	0.291***

*** Represents *p*-value ≤0.05.

the COVID-19 pandemic crisis as an event. Table 2 showcases the results of the association of CEOs' different orientations with SM reputation of CEO's sustainability posts on their Twitter handles before and during the pandemic crisis.

From the results of multiple linear regression, it is evident that there exists a statistical difference between the Fortune CEO's different strategic orientations on social media on their SM reputation before and during the COVID-19 pandemic crisis, with *p*-value >0.05 and R^2 values 0.155 (before the crisis) and 0.291 (after the crisis). There is a significant decline in the β value of CEO's SDG orientation with the CEO SM reputation with value changing from 0.413 (before the crisis) to 0.283 (after the crisis). Also, the CEO's EO is found to have no significant effect on SM reputation before the crisis and has a significant positive effect on SM reputation with β value 0.196 during the crisis. However, there is no significant relationship found between the CEO's leadership style in their SM communication surrounding sustainability and their SM reputation both before and during the COVID-19 pandemic crisis. Hence, we reject hypothesis H1c and accept hypothesis H1b and H1c.

5.4. Event study 2: CEO's social media reputation and firm performance

Multiple linear regression results showcase that there is a significant change in the association of the CEO's SM reputation with the firm financial performance before and during the pandemic crisis. Table 3 summarizes the association between the CEOs' SM reputation and firm financial performance.

It is evident from the results (Table 3) that there is a positive association between the CEOs' SM reputation with the firm financial performance. The results indicate that there is no change in the association of the CEO's SM reputation and RoA with the p-value >0.05. A significant difference in the CEO's SM reputation with Sales revenue is found with a p-value >0.05 before the COVID-19 pandemic and a p-value of ≤0.05 during the COVID-19 pandemic. Also, the relationship between the CEO's SM reputation and sales revenue is positive, with a 23.280 β value and 0.793 R^2 value. The change in the relationship of the CEO's SM reputation with TobinsQ before and during the COVID-19 pandemic is also significant, with a p-value >0.05 before the pandemic and a p-value of ≤0.10 during the COVID-19 pandemic. There is a positive relationship between the CEO's SM reputation and TobinsQ during the COVID-19 pandemic with a 0.983 β value and 0.433 R^2 value. Hence, we reject hypothesis H3a and accept hypotheses H3b and H3c.

The result showcases that the relationship of CEOs' SM reputation with firm financial parameters is not significant before the COVID-19 pandemic crisis, while it is significant during the COVID-19 pandemic crisis. Hence, the findings show a significant change in the association of CEOs' SM reputation with firm financial performance during the pandemic crisis. Table 4 summarizes the hypotheses and results of this study.

Table 3
Multiple regression analysis results of CEOs' social media reputation and firm financial performance.

Independent variables	Dependent variables		
	Firm performance		
	RoA	Sales revenue	TobinsQ
Before-Crisis			
CEO_SM_Reputation	−0.001	4.926	0.425**
R^2	0.133	0.437	0.283
During-Crisis			
CEO_SM_Reputation	−0.075	23.280***	0.983***
R^2	0.132	0.793***	0.433**
Control Variables			
Firm Size	Controlled		
Sector	Controlled		

*** Represents p-value ≤0.05.
** Represents p-value ≤0.10.

6. Discussion

CEOs' effective communication on SM regarding sustainability is crucial to cultivating meaningful engagement with the stakeholders by shaping the information environment and influencing firm performance (Jung et al., 2018; Heavey et al., 2020; Gruber, 2023). Building on the Stakeholder theory and RBV theory, we elaborate on how different orientations of the CEOs in their SM disclosure shape their information disclosure decisions for enhanced stakeholder engagement, influencing their SM reputation and having further implications on firm performance. The study analysis reveals the importance of change in strategic utilization of SM for the CEO communication through their different orientations for influencing the firm performance during the crisis.

Firstly, our studies aim to identify the change in different strategic orientations of CEOs as a strategy to alleviate the adverse effects of the crisis. Our findings indicate a change in CEOs' strategic orientations in their SM narratives before and during the COVID-19 pandemic. The CEOs' SDG orientation and leadership style are changing significantly, whereas the change in their EO is found to be insignificant. The study findings are aligned with the previous studies that showcase evidence of firms' changing orientation towards different sustainability dimensions after the onset of the COVID-19 pandemic crisis (Barreiro-Gen et al., 2020). Fischer and Reuber (2014) in their research highlight the importance of EO to capitalize on emerging aspects while engaging in sustainability initiatives. However, based on our research results, it appears that CEOs fall short in integrating EO into their effective communication on SM to tackle the issues arising from the crisis. Further, this study complements the other studies where the important role of CEOs' leadership style in addressing the challenges of information asymmetry among the stakeholders has been highlighted (Varma, 2021).

Second, using stakeholder theory as a theoretical lens, we identify

Table 4
Hypotheses summary.

Hypotheses summary			
S. No.	Hypothesis (Change in effect before and during the crisis)	Proposed effect	Result
H1a	CEO's SDG orientation	NA	Accepted
H1b	CEO's EO	NA	Accepted
H1c	CEO's leadership style	NA	Accepted
H2a	CEO's SDG orientation ➔ CEO SM reputation	+	Accepted
H2b	CEO's EO ➔ CEO SM reputation	+	Accepted
H2c	CEO's leadership style ➔ CEO SM reputation	Not Significant	Rejected
H3a	Stakeholder endorsement on SM ➔ Firm's return on assets (RoA)	Not Significant	Rejected
H3b	Stakeholder advocacy on SM ➔ Firm's sales revenue	+	Accepted
H3c	Stakeholder endorsement on SM ➔ Firm's TobinsQ	+	Accepted

whether the change in CEO's different orientations in their SM communication around sustainability during the crisis has any influence on their SM reputation. The CEO's SDG orientation leads to a decline in stakeholder engagement and SM reputation during the crisis. The CEO's EO leads to a higher engagement and, hence, a positive SM reputation during the crisis. The change in the CEO's leadership style during the COVID-19 pandemic crisis is found to have no significant impact on stakeholder engagement and their reputations. Therefore, it is evident that CEOs change their orientations in their SM communication during a crisis, and it further has implications for their stakeholder engagement and reputation (Yao, 2022; Varma, 2021; Heavey et al., 2020).

Further, this study explores the change in the relationship between the CEOs' SM reputation and firm financial performance during the COVID-19 pandemic crisis by utilizing RBV theory as a theoretical lens. The findings reveal a significant impact of CEOs' SM reputation on the firm performance i.e. sales revenue and TobinsQ during the COVID-19 pandemic crisis. It is observed that the relationship between the CEO's SM reputation with sales revenue and TobinsQ is not significant before the pandemic crisis, while it is positively significant during the crisis. The results, therefore, contribute to the existing literature where an overall positive association of the CEO's SM reputation is found with the firm financial performance during the crisis (Pfarrer et al., 2008).

6.1. Theoretical contributions

Our research carries several theoretical implications. First, in light of sustainability, there has been an increased need and interest in promoting sustainability initiatives during a crisis. The increased demand for information disclosures surrounding sustainable practices adopted by firms due to the COVID-19 pandemic has been observed recently (Jha and Verma, 2022; Gruber, 2023). Leaders utilize SM platforms to highlight the firm's initiatives towards different sustainability practices (Khizar et al., 2022; Buck et al., 2022; Turan, 2022). The study findings showcase a shift from the inclination of CEO tweets from economic and social SDGs to all three aspects of sustainability, i.e. economic, environmental, and social, during the COVID-19 pandemic crisis, thus providing a clear signal that firms and their CEOs change their different sustainability aspects adoption as per the business and societal demands.

Second, earlier research either focused on market-oriented strategies or community engagement use of SM (Trainor et al., 2014; Men and Tsai, 2016; Khizar et al., 2022). The COVID-19 pandemic made it important for the firms and their leaders to effectively utilize SM platforms to connect with the stakeholders (Banks et al., 2022) while highlighting the firm's initiatives towards sustainable business practices (Khizar et al., 2022; Buck et al., 2022; Turan, 2022). Higher stakeholder engagement on the CEO's Twitter posts by liking or sharing their tweets is symbolic of higher CEO reputation on SM (De Luca et al., 2022; Attié and Meyer-Waarden, 2022; Grover et al., 2019). Therefore, this study adds value to the current academic literature on the opportunity for CEOs to effectively leverage SM through different orientations (SDG, entrepreneurial, and leadership style) in their sustainability communication on SM to influence their engagement with stakeholders during a crisis. The findings add to stakeholder theory applications by providing significant insights into important implications of change in CEOs' different orientations in their information disclosures on SM platforms on stakeholder engagement levels and, in turn, CEO reputation.

Third, RBV theory has been used to theorize between the SM reputation built through strategic orientations of the CEO in their SM communication during crisis and the firm financial performance. Reputation has been studied as a key effort to understand why some organizations outperform others and has been linked to positive organizational performance (Shamsie, 2003; Boyd et al., 2010). We argue that the SM reputation built during a crisis acts as a strategic resource. A strong SM reputation may bring customer loyalty and trust, and attract customers who align with sustainability values, leading to increasing sales revenue (Mention et al., 2019). SM reputation of the CEO built through unique orientations (SDG, entrepreneurial, and leadership styles) in their sustainability-related posts may create positive sentiment in investors who value sustainability and responsible business practices hence higher long-term-price-to-earnings-ratio (Flammer and Bansal, 2017). Therefore, the study contributes to the expansion of applications of RBV theory by exploring CEOs' different orientations in their SM communication and their reputation acting as a unique and intangible yet critical resource for the firms resulting in the competitive advantage for achieving higher firm performance during crisis.

6.2. Practical implications

This study offers interesting insights into the role of CEO orientations in their SM communication surrounding sustainability for enhanced firm performance. It has multiple implications for practice and corporate strategy to be adopted by the firms and their CEOs. First, with the UN Agenda 2030 and the COVID-19 pandemic crisis, it became crucial for the firms to focus more on sustainability disclosures. Thus selection of topics that are relevant and interesting at the same in the CEO's SM communication results in better stakeholder engagement and enhanced reputation. Second, it is crucial to understand how CEOs adapt different orientations in their SM communication during a crisis. The study showcases how the shift in orientation in CEOs' communication can help mitigate the crisis effects on organizational performance. The SM communication data can guide the CEOs on the dos and don'ts' of crisis communication, potentially saving them from massive reputational damage and financial losses. Third, the study findings clearly indicate that CEOs' strategic orientations in their communication on SM directly impacts their SM reputation. Failure to strategically adapt different orientations, especially during a crisis can result in reputational risks. Hence, the top management leaders can use the insights from the study to continuously shape their SM persona considering the direct implications on their reputation. Fourth, through the empirical data, this study makes it apparent how the CEOs' reputation built on SM through their sustainability disclosures influences firm performance during the changing environment, for instance, the COVID-19 pandemic crisis. Lastly, the CEO's disclosures must be consistent with the firm's mission and values which can positively impact the stakeholder trust and financial performance.

6.3. Limitations and future research work

Current research investigates how SM can be leveraged as a resource by the CEOs of organizations for the purpose of better stakeholder engagement, building reputation, and influencing firm performance. The current study uses Twitter posts of the CEOs through an event study. Similarly, other SM platforms can also be studied to understand the role of different orientations in the communication adopted by the leaders on various platforms like Facebook and further identify which SM is the most effective communication means for engaging with the stakeholders during a crisis. Also, we focus only on sustainability tweets posted by the CEOs, future research can explore different topics of relevance. The current study has been limited to only the textual features of SM posts, which can further be extended by utilizing the visual features of these posts. Further, to get a more holistic view of the SM use by firms for crisis management, a further investigation shall be done in contexts of different crisis types. Therefore, the above limitations highlighted may be explored as future research directions for improving organizations' current theoretical understanding of strategic communication styles as a crisis management strategy.

7. Conclusions

The study significantly contributes to the understanding of effective communication through CEO's different orientations on SM, especially in times of crisis. Theoretically, it adds to the stakeholder theory by

focusing on how CEO's changing orientations in SM communication cater to the changing informational needs of their stakeholders over time, leading to higher stakeholder engagement and reputation. It enriches the RBV theory by detailing how CEO reputation on SM can be a unique competitive resource for the firms to achieve better financial performance for the firm. The study provides a better understanding and actionable guidance on mitigating damage to CEO reputation and firm performance loss during a crisis. The study also highlights that SM now plays a vital role in the successful stakeholder engagement and in influencing the firm performance even during turbulent times of crisis, thereby, contributing to various research avenues such as strategic utilization of SM, role of CEOs' individual orientations for enhanced financial performance, the social psychology of the online stakeholders for their decision making, and crisis management.

CRediT authorship contribution statement

Hitesha Yadav: Methodology, Writing – original draft. **Arpan Kumar Kar:** Supervision, Writing – original draft, Writing – review & editing. **Smita Kashiramka:** Formal analysis, Methodology, Supervision, Writing – review & editing. **Nripendra P. Rana:** Conceptualization, Investigation, Validation, Writing – review & editing.

Declaration of competing interest

There is no conflict of interest.

Data availability

Data will be made available on request.

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